PROGRAM 9 – TRAVELLING SALESMAN PROBLEM (TSP)

AIM

To write a Python program to implement the Travelling Salesman Problem using permutations and find the minimum cost tour.

ALGORITHM

1. Start the program.
2. Define the cities and their distances.
3. Read the starting city from the user.
4. Generate all possible routes between the remaining cities.
5. Calculate the total distance for each route.
6. Add the distance required to return to the starting city.
7. Compare all route costs.
8. Select the route with the minimum cost.
9. Display the best route and minimum distance.

CODE

```
from itertools import permutations

cities=["A","B","C","D"]

distance={"A":{"A":0,"B":10,"C":15,"D":20},"B":{"A":10,"B":0,"C":35,"D":25},"C":{"A":15,"B":35,"C":0,
"D":30},"D":{"A":20,"B":25,"C":30,"D":0}}

start=input("Enter starting city: ").upper()

if start not in cities:
    print("Invalid city")
else:
    others=[city for city in cities if city!=start]
    min_cost=float("inf")
    best_route=None
    for route in permutations(others):
        path=(start,)+route+(start,)
        cost=sum(distance[path[i]][path[i+1]] for i in range(len(path)-1))
        if cost<min_cost:
            min_cost=cost
            best_route=path
    print("Best Route:"," -> ".join(best_route))
```

```
print("Minimum Distance:",min_cost)
```

INPUT

```
Enter starting city: A
```

OUTPUT

```
Best Route: A -> B -> D -> C -> A
Minimum Distance: 80
```

RESULT

Thus, the Travelling Salesman Problem was successfully implemented in Python to find the minimum cost route.